Simulating Swarm Robots for Space Stations

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***Abstract*—This paper investigates the application of biologically inspired swarm robotics algorithms within confined microgravity environments representative of the interior of future space stations. A browser-based simulation environment was developed to model free-floating robotic agents operating inside a confined orbital habitat inspired by the TESSERAE modular robotic architecture. The simulation utilizes local interaction laws outlined in previous swarm robotics research inspired by schools of fish. The project also uncovers how there is a relationship between underwater-inspired swarm behaviors and space robotics. In addition to the simulation framework, this work proposes a conceptual robotic platform capable of physically reproducing the simulated behaviors through omnidirectional propulsion and local sensing. The simulation serves both as a platform for analyzing swarm behaviors and as a tool for investigating future applications of intra-vehicular robotic swarms. This work contributes to ongoing research in swarm robotics, bio-inspired autonomous systems, and microgravity robotics by exploring how decentralized collective behaviors may be adapted for confined space environments.**

I. Introduction

Underwater and space technology research have long been aligned due to some of the similarities in problems that occur in extreme environments. Multi-robot systems have been a large area for research in both autonomous underwater systems and for space technology applications. This project is inspired by both those fields of research. Additionally, zooming into a subsection of multi-robot systems we will find swarm robotics. Inspired by collective biological systems such as schools of fish, flocks of birds, and colonies of insects, swarm robotic systems mainly utilize decentralized interactions between agents to produce coordinated large-scale behavior. This project investigates how bio-inspired swarm algorithms, particularly those modeled after schools of fish, can be applied within microgravity environments for the interior of space stations.

The primary objective of this project is to create a three-dimensional simulation of a robotic swarm operating inside the TESSERAE which is a confined space station environment and to evaluate the behaviors produced by previously proposed swarm algorithms [1]. The simulation focuses on free-floating robotic agents in microgravity and investigates how biologically inspired interaction rules influence aggregation, dispersion, and milling behaviors. These behaviors are based primarily on the work presented in the paper titled "Implicit coordination for 3D underwater collective behaviors in a fish-inspired robot swarm", which demonstrates fish-school-inspired coordination strategies capable of producing emergent collective motion through local interactions between agents [2]. In particular, this project focuses on three major behaviors explored in that work: aggregation, dispersion, and milling.

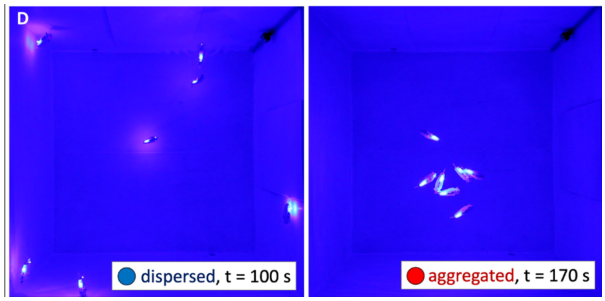


Fig. 1. The Fish robots from [2] showing aggregation and dispersion behavior.

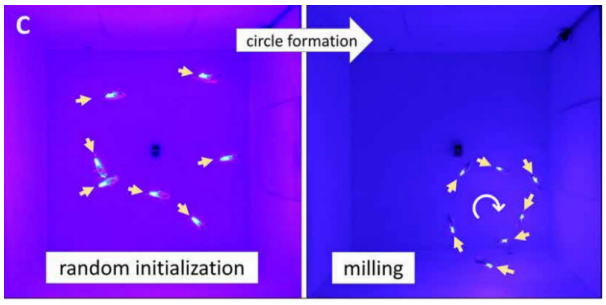


Fig. 2. The Fish robots from [2] showing milling behavior.

The space station environment utilizes the design of the TESSERAE as the external boundaries. This was chosen because the TESSERAE also has multi-agent ties due to its proposed modularity when launched. Space stations like the TESSERAE are also possible for future use, so this project is quite forward looking, especially with the recent push for more space exploration. Employing the TESSERAE structure and the school of fish based algorithms in this simulation based display how complex interactions between agents and their environment can result from relatively simple behavior.

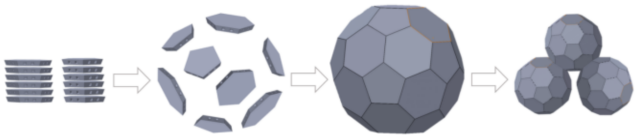


Fig. 3. The TESSERAE self-assembly process. Starting with flat tiles and ending with several stations coming together.

II. Simulation Development and Resutls

In this section I will outline the process of the simulation development as well as describing several of the choices made during the process. The simulation can be run by users if you get use the link from my github (linked at the end of this paper). The results of the simulation include a three-dimension microgravity simulation which has several adjustable parameters. Some of the parameters are adjustable within the body of the simulation itself whereas others are adjustable through the preconditions in the code. The parameters on the simulation itself include being apply to apply forces in the x, y, or z direction onto all of the agents. It also includes the parameters that cause the aggregation and dispersion behavior (epsilon and sigma). Lastly the simulation contains the parameters for the milling behavior based on vision. The code parameters mainly include being able to adjust the number of agents.

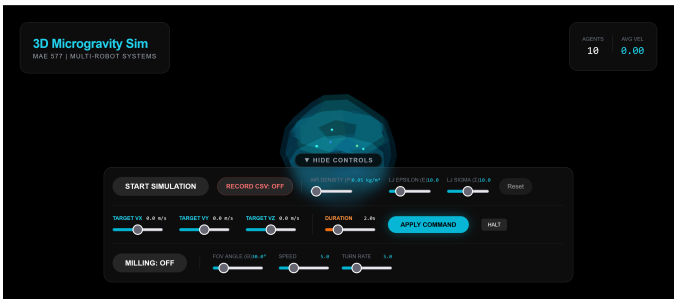


Fig. 4. The simulation when it is first opened.

The parameters can be toggled down which allows for better viewing of the simulation itself. Additionally after each start and stop of the simulation you can also denote if you would like to save a csv file for the results of the simulation. This file includes the x-y-z position of each of the agents and their velocity in each of those directions as well.

There are also rooms within the TESSERAE and those also act as walls for the robots unless they go through the door in the middle of the section. For example in this image the robots get stuck at the top of each room in each section when a force in the z direction occurs.

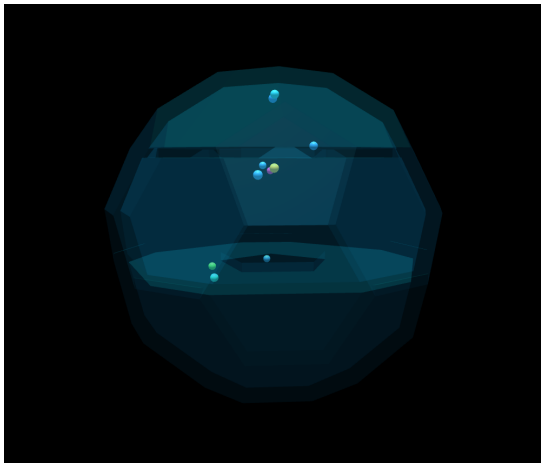


Fig. 5. simulation after force in Z direction.

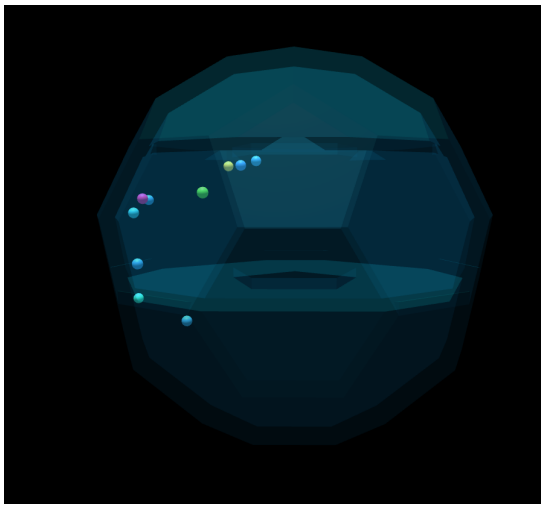


Fig. 6. simulation after force in X direction.

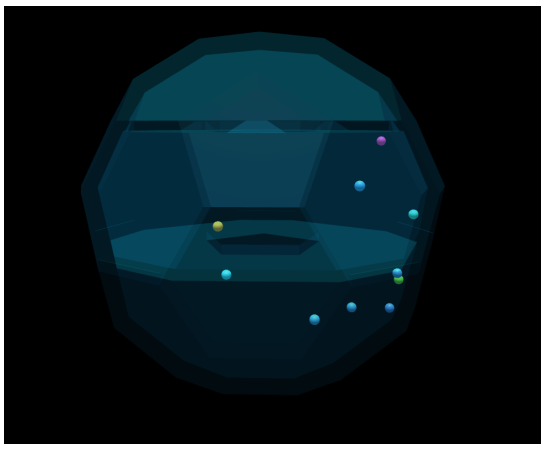


Fig. 7. simulation after force in Y direction.

The swarm interactions implemented within the simulation are partially based on the Lennard-Jones potential, a mathematical model commonly used in molecular dynamics and

particle simulations to represent attraction and repulsion forces between particles [3]. This is the same principle used in the Berlinger Et Al. paper [2]. In this project, the Lennard-Jones interaction law is adapted to model local interactions between robotic agents in order to produce emergent aggregation and spacing behaviors. The Lennard-Jones potential is defined as

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right] \tag{1}$$

where r represents the distance between two agents, ϵ controls the strength of the interaction, and σ defines the characteristic interaction distance between agents. The repulsive term, proportional to r^{-12}, becomes dominant when agents are very close together and prevents collisions or overlap between robots. The attractive term, proportional to r^{-6}, acts over larger distances and encourages agents to remain cohesive as a swarm. Within the simulation, the parameter σ effectively controls the preferred spacing between robotic agents, while ϵ determines how strongly agents attract and repel one another. Larger values of ϵ produce stronger cohesive behavior and more aggressive collision avoidance, whereas larger values of σ increase the equilibrium spacing between agents and create more dispersed swarm formations. By tuning these parameters, different collective behaviors such as aggregation, dispersion, and stable milling patterns can be produced within the confined microgravity environment.

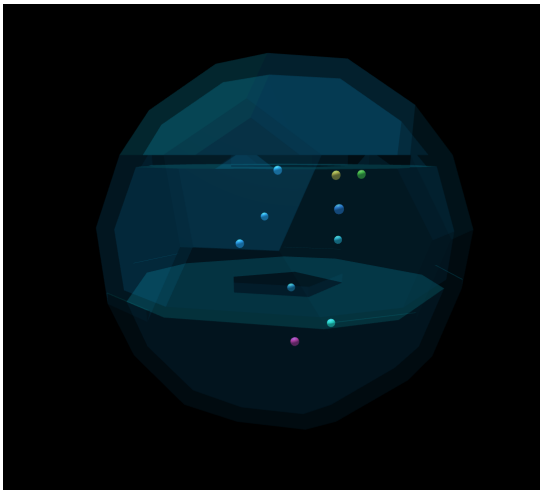


Fig. 8. Aggregation of the agents

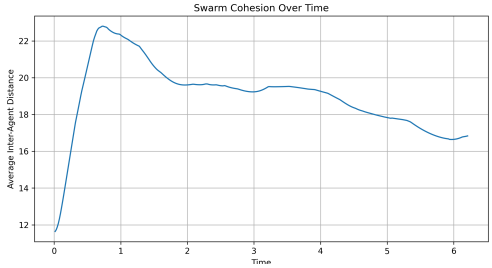


Fig. 9. graph showing the average distance between the agents

As we can see the average distance between the agents is decreasing as time goes on showing that the agents are infact aggregating. Even though it is not super clear through the simulation.

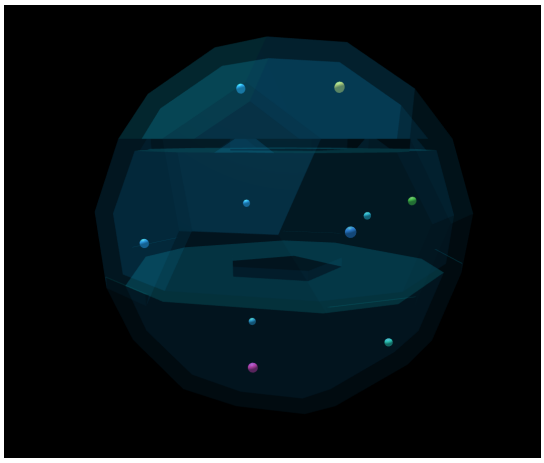


Fig. 10. Dispersion in Simulation

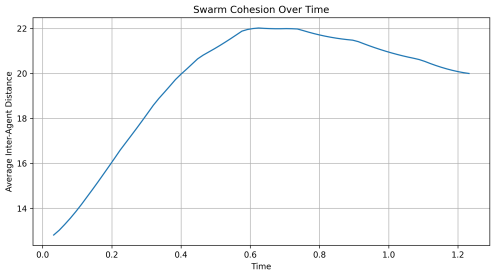


Fig. 11. graph showing the average distance between the agents.

This shows that the distance between agents increases over time. The reason why it begins tapering off at the end is that there is a max distance the agents can be apart before they hit the wall and collide back towards the center.

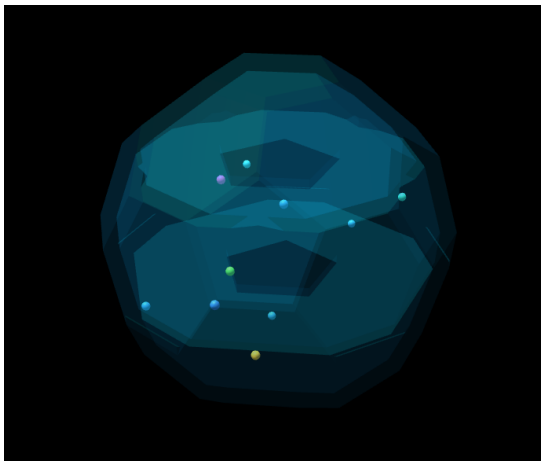


Fig. 12. this shows the milling behavior.

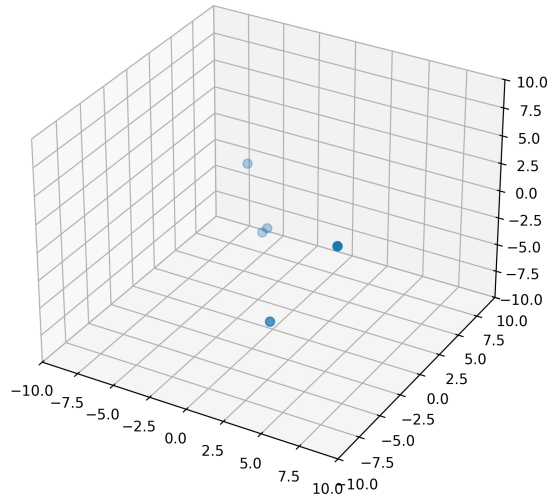


Fig. 13. 5 agents in the 3D matlplotlib

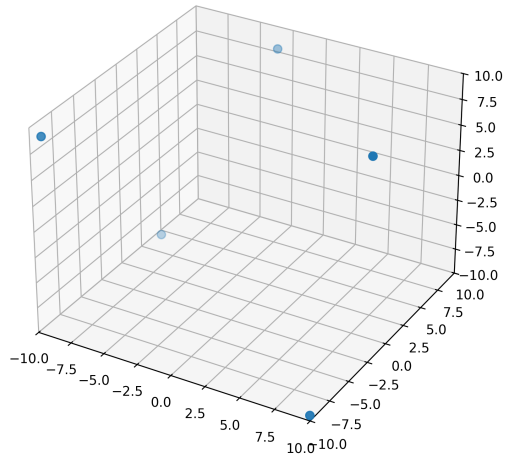


Fig. 14. 3D matplotlib showing dispersion

The milling behavior is a bit more difficult to see from a still image but it is more clear when the simulation is running.

A. Methods

Before I began the browser-based simulation environment for the milling. I did some preliminary swarm behavior testing with Python and Matplotlib's three-dimensional plotting tools. Although the visualization methods differed, much of the underlying simulation logic remained similar between the Python and browser implementations. This was a simple way to test the functionality before embedding it in the full simulation.

Then when moving on to the browser based version using javascript and Three.js. The simulation development process began with having a simple cubic environment that representing the interior of a confined space station module. Robotic agents were then initialized inside the cube and allowed to move freely in three dimensions. I did this to confirm the wall collisions and also to see how the agents would move around in microgravity.

Three.js is a javascript graphics library that is used for three-dimensional rendering and visualization. A browser-based framework was selected for several reasons. First, it allowed for significantly greater flexibility in developing interactive user interface components such as sliders, buttons, and parameter controls. Second, the browser environment improved accessibility, allowing the simulator to be more easily shared and used across different systems without requiring specialized software installation. Finally, the separation between backend physics calculations and frontend visualization simplified experimentation and debugging.

The simulator architecture was designed such that much of the physics computation and swarm logic could be handled separately from the graphical rendering system. Python was initially used for prototyping and numerical experimentation, while JavaScript and three.js were used primarily for visualization and interactive control in the browser implementation. This separation allowed the project to combine the computational flexibility of Python with the visualization capabilities of modern web-based graphics frameworks.

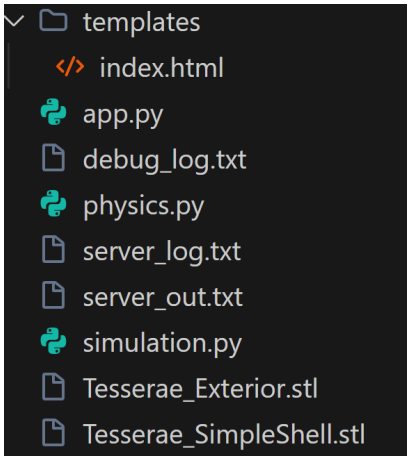


Fig. 15. The file tree showing the backend python based physics and motion control.

The simulation.py file contains the default parameters that the simulation begins with. The physics.py file contains all the physics elements such as the drag due to the amount of air in the space station as well as the aggregation, dispersion, and milling behaviors.

The three.js library was particularly useful due to its support for real-time three-dimensional rendering and its compatibility with WebGL. Unlike my previous projects and experience where I relied heavily on two-dimensional frameworks such as Pygame, the browser-based three-dimensional environment allowed for a more natural representation of free-floating robotic agents in microgravity and the rendering of the TESSERAE system.

B. Challenges and Limitations

During development, several technical challenges and limitations emerged. One of the primary difficulties involved accurately modeling physical interactions between agents and the surrounding environment. In particular, drag forces, collision handling, and wall interactions proved difficult to tune realistically while maintaining stable simulation performance. Because agents operate in a microgravity-inspired environment, small errors in force calculations or numerical integration could quickly produce unstable or unrealistic motion.

Another challenge involved integrating orbital and microgravity-inspired motion models into a browser-based framework. Existing JavaScript libraries for orbital control and physical simulation were often difficult to adapt to the swarm robotics context and occasionally introduced instability or visual artifacts into the simulation. Additionally, the real-time nature of the browser visualization sometimes resulted in graphical glitches or inconsistent rendering behavior, particularly when large interaction forces were introduced between agents.

To assist with development, Google Antigravity was used as an integrated development environment that incorporates generative AI tools for web application and interface development. The platform was useful for accelerating portions of the user interface and visualization development process, particularly during early prototyping stages. However, while these tools improved development speed, they also introduced challenges related to debugging, code reliability, and maintaining consistent system behavior. In many cases, manually modifying or restructuring generated code was necessary to ensure simulation stability and maintainability. The use of AI-assisted development tools also raised broader questions regarding their role in engineering workflows and whether such systems may influence creativity or design approaches during development.

Despite these challenges, the resulting simulator provides a flexible platform for exploring swarm robotic behavior in confined microgravity-inspired environments and serves as a foundation for future experimentation involving more advanced physics, communication systems, and autonomous coordination algorithms.

III. Design Proposal

In addition to the simulation and swarm algorithm development, this project also proposes a conceptual robotic platform capable of physically reproducing the behaviors explored within the simulation environment. Since the simulated robots operate in a confined microgravity environment and are required to perform precise three-dimensional motion, the robotic design must prioritize omnidirectional maneuverability, localization accuracy, and collision tolerance.

One of the primary design considerations is the propulsion and actuation system. In order to achieve controlled motion in microgravity, the proposed platform would likely require approximately six independently controllable actuators or propellers. This design choice is motivated by prior intravehicular robotic systems proposed in the literature, which utilize multiple thrusters to enable precise translational and rotational control in confined orbital environments [4]. A six-actuator configuration would allow the robot to generate forces and torques across multiple axes, enabling stable maneuvering, orientation control, and fine positional adjustments while floating inside a space station.

The robotic platform would also require onboard sensing systems capable of detecting neighboring agents and the surrounding environment. Since swarm coordination algorithms rely heavily on local interactions between nearby robots, each agent must be capable of estimating the relative position and motion of neighboring robots. One proposed solution is the use of vision-based localization and sensing systems, such as onboard cameras or optical tracking methods. Vision-based approaches are particularly attractive in this context because localization remains one of the major challenges in autonomous robotics, especially in environments where external positioning systems may be limited.

Traditional robotic localization methods frequently rely on inertial measurement units (IMUs) to estimate position, orientation, and angular velocity. However, IMU-based localization systems are prone to cumulative drift over time, as small measurement errors continuously accumulate during long-duration operation. In confined microgravity environments, where precise positioning and relative coordination are essential, this drift can significantly reduce localization accuracy. As a result, vision-based interaction and sensing systems may provide a more reliable alternative for swarm coordination by allowing robots to directly observe neighboring agents and environmental features. This aligns well with the biologically inspired interaction models explored in this project, where agents primarily respond to locally observed neighbors rather than relying on centralized positioning systems.

Another major design requirement is collision resilience. Since the robots are intended to operate in close proximity to one another and within the interior of a space station, the platform must be capable of safely handling repeated low-speed collisions without damaging either the robots themselves or the surrounding station infrastructure. Potential design considerations include soft exterior materials, compliant structures, protective cages surrounding propulsion systems, or energy-absorbing bumpers. These features would help ensure safe operation during aggregation and milling behaviors where agents intentionally move in dense formations.

While this robotic platform proposed in this work remains conceptual, the design serves as a useful framework for connecting the simulation environment to potential future physical implementations.

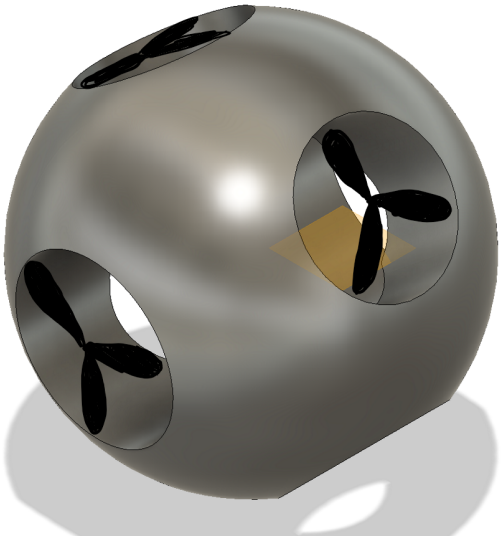


Fig. 16. Proposed design for the robots in the simulation.

IV. Conclusion

This project explored the application of biologically inspired swarm robotics algorithms within confined microgravity environments representative of the interiors of future orbital habitats. By combining concepts from swarm robotics, collective animal behavior, and intra-vehicular space systems, a three-dimensional simulation environment was developed to investigate how decentralized robotic agents may coordinate and self-organize in microgravity. Drawing inspiration from fish-schooling behaviors, the simulation demonstrated emergent collective behaviors including aggregation, dispersion, and milling through the use of local interaction laws and Lennard-Jones-inspired attraction and repulsion forces.

V. Future Work

There is a lot of additions that could be made to this simulator. There are a lot of complexities to explore in regards to how the flexibility of the platform and web-based browser can be leveraged for further testing of different algorithms. I also read a pretty new paper detailing some of the alternatives to the Lennard-Jones potential field in the field of physical sciences [**?**]. They discuss simplifications of the Lennard-Jones potential which would allow for easier calculations mathematically in the simulation. In the TESSERAE paper there is a discussion of connecting multiple sections together to create a larger station. Including this aspect in the simulation and being able to construct largest TESSERAE structures is also an interesting avenue to explore. Building the proposed physical multi-agent robot to conduct the motion that the simulation shows could be aspect to explore in the future. Another aspect that would be interesting to explore is how space algorithms and tactics employed can be applied to underwater robots. Especially when thinking about multi-agent systems, many of which exist both in space and underwater.

Acknowledgments: Thank you Professor Radhika Nagpal for teaching this course and encouragement throughout the semester. Thank you to Sneha Ramshanker for help brainstorming during the early stages and exploration of this project. Thank you to all my peers in the class for their amazing presentations, I learned so much and I hope I showed even a fraction of that through this research project.

AI Use: AI (gemini and ChatGPT) was used for coding parts of the simulation and for syntax on how to write some python code. The analysis and writing for this paper is all my own. The code can be found here: https://github.com/yushrag/MAE577_finalproject. To run the simulation you can download the code and then run python app.py.

The formatting of this paper comes from an IEEE submitted paper [5]

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